Fifth Degree Function Report

Polynomial

$$f(q) = q^2 \left(aq^3 + bq^2 + cq + d\right)$$

Naive

$$g_{\text{naive}}(X) = X^2 \left(X^3 a + X^2 b + Xc + d\right)$$

$$\begin{aligned}\mathbb{E}[g_{\text{naive}}(X)] = aq^5 + bq^4 + 24\frac{\Delta}{\epsilon}^4 (5aq + b) \\ + 2\frac{\Delta}{\epsilon}^2 \left(10aq^3 + 6bq^2 + 3cq + d\right) + cq^3 + dq^2\end{aligned}$$

$$\begin{aligned}\operatorname{Var}(g_{\text{naive}}(X)) = 2\frac{\Delta}{\epsilon}^2 \Bigg(1814400a^2\frac{\Delta}{\epsilon}^8 \\ + 288\frac{\Delta}{\epsilon}^6 \left(3125a^2q^2 + 1250abq + 140ac + 69b^2\right) \\ + 24\frac{\Delta}{\epsilon}^4 \left(3050a^2q^4 + 2440abq^3 + 810acq^2 + 200adq + 408b^2q^2 \right. \\ \left. + 204bcq + 28bd + 15c^2\right) \\ + 2\frac{\Delta}{\epsilon}^2 \left(1100a^2q^6 + 1320abq^5 + 720acq^4 + 340adq^3 + 372b^2q^4 \right. \\ \left. + 372bcq^3 + 156bdq^2 + 81c^2q^2 + 54cdq + 5d^2\right) \\ + q^2 \left(25a^2q^6 + 40abq^5 + 30acq^4 + 20adq^3 + 16b^2q^4 + 24bcq^3 \right. \\ \left. + 16bdq^2 + 9c^2q^2 + 12cdq + 4d^2\right)\Bigg)\end{aligned}$$

$$\begin{aligned}\operatorname{MSE}(g_{\text{naive}}) = 2\frac{\Delta}{\epsilon}^2 \Bigg(1814400a^2\frac{\Delta}{\epsilon}^8 + 20160\frac{\Delta}{\epsilon}^6 \left(45a^2q^2 + 18abq + 2ac + b^2\right) \\ + 360\frac{\Delta}{\epsilon}^4 \left(210a^2q^4 + 168abq^3 + 56acq^2 + 14adq + 28b^2q^2 \right. \\ \left. + 14bcq + 2bd + c^2\right) + 12\frac{\Delta}{\epsilon}^2 \left(200a^2q^6 + 240abq^5 + 130acq^4 \right. \\ \left. + 60adq^3 + 68b^2q^4 + 68bcq^3 + 28bdq^2 + 15c^2q^2 + 10cdq + d^2\right) \\ + q^2 \left(25a^2q^6 + 40abq^5 + 30acq^4 + 20adq^3 + 16b^2q^4 + 24bcq^3 \right. \\ \left. + 16bdq^2 + 9c^2q^2 + 12cdq + 4d^2\right)\Bigg)\end{aligned}$$

$$\operatorname{Bias}(g_{\text{naive}}) = 2\frac{\Delta}{\epsilon}^2 \left(10aq^3 + 6bq^2 + 12\frac{\Delta}{\epsilon}^2 (5aq + b) + 3cq + d\right)$$

Unbiased

$$g_{\text{unb}}(X) = X^5a + X^4b + X^3c + X^2d - 2\frac{\Delta}{\epsilon}^{2}\left(10X^3a + 6X^2b + 3Xc + d\right)$$

$$\mathbb{E}[g_{\text{unb}}(X)] = q^2\left(aq^3 + bq^2 + cq + d\right)$$

$$\begin{aligned}\operatorname{Var}(g_{\text{unb}}(X)) = 2\frac{\Delta}{\epsilon}^{2}\Bigg(&1152000a^2\frac{\Delta}{\epsilon}^{8} + 288\frac{\Delta}{\epsilon}^{6}\left(2000a^2q^2 + 800abq + 85ac + 46b^2\right) \\ &+ 12\frac{\Delta}{\epsilon}^{4}\left(4000a^2q^4 + 3200abq^3 + 1020acq^2 + 220adq + 552b^2q^2\right. \\ &\left.+ 276bcq + 36bd + 21c^2\right) + 2\frac{\Delta}{\epsilon}^{2}\left(800a^2q^6 + 960abq^5 + 510acq^4\right. \\ &\left.+ 220adq^3 + 276b^2q^4 + 276bcq^3 + 108bdq^2 + 63c^2q^2 + 42cdq + 5d^2\right) \\ &+ q^2\left(25a^2q^6 + 40abq^5 + 30acq^4 + 20adq^3 + 16b^2q^4 + 24bcq^3\right. \\ &\left.+ 16bdq^2 + 9c^2q^2 + 12cdq + 4d^2\right)\Bigg)\end{aligned}$$

$$\begin{aligned}\operatorname{MSE}(g_{\text{unb}}) = 2\frac{\Delta}{\epsilon}^{2}\Bigg(&1152000a^2\frac{\Delta}{\epsilon}^{8} + 288\frac{\Delta}{\epsilon}^{6}\left(2000a^2q^2 + 800abq + 85ac + 46b^2\right) \\ &+ 12\frac{\Delta}{\epsilon}^{4}\left(4000a^2q^4 + 3200abq^3 + 1020acq^2 + 220adq + 552b^2q^2\right. \\ &\left.+ 276bcq + 36bd + 21c^2\right) + 2\frac{\Delta}{\epsilon}^{2}\left(800a^2q^6 + 960abq^5 + 510acq^4\right. \\ &\left.+ 220adq^3 + 276b^2q^4 + 276bcq^3 + 108bdq^2 + 63c^2q^2 + 42cdq + 5d^2\right) \\ &+ q^2\left(25a^2q^6 + 40abq^5 + 30acq^4 + 20adq^3 + 16b^2q^4 + 24bcq^3\right. \\ &\left.+ 16bdq^2 + 9c^2q^2 + 12cdq + 4d^2\right)\Bigg)\end{aligned}$$

Comparison

$$\text{mean gap} = -2\frac{\Delta}{\epsilon}^{2}\left(10aq^3 + 6bq^2 + 12\frac{\Delta}{\epsilon}^{2}\left(5aq + b\right) + 3cq + d\right)$$

$$\begin{aligned}\text{variance gap} = -24\frac{\Delta}{\epsilon}^{4}\Bigg(&55200a^2\frac{\Delta}{\epsilon}^{6} + 24\frac{\Delta}{\epsilon}^{4}\left(1125a^2q^2 + 450abq + 55ac + 23b^2\right) \\ &+ \frac{\Delta}{\epsilon}^{2}\left(2100a^2q^4 + 1680abq^3 + 600acq^2 + 180adq + 264b^2q^2\right. \\ &\left.+ 132bcq + 20bd + 9c^2\right) + q\left(50a^2q^5 + 60abq^4 + 35acq^3 + 20adq^2\right. \\ &\left.+ 16b^2q^3 + 16bcq^2 + 8bdq + 3c^2q + 2cd\right)\Bigg)\end{aligned}$$

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\begin{aligned}
\text{MSE gap} = -4\frac{\Delta}{\epsilon}^4 \Bigg(& 331200a^2\frac{\Delta}{\epsilon}^6 + 400a^2q^6 + 480abq^5 + 270acq^4 + 140adq^3 \\
& + 132b^2q^4 + 132bcq^3 + 60bdq^2 \\
& + 144\frac{\Delta}{\epsilon}^4 \left(1150a^2q^2 + 460abq + 55ac + 24b^2\right) + 6\frac{\Delta}{\epsilon}^2 \left(2300a^2q^4\right. \\
& \left. + 1840abq^3 + 660acq^2 + 200adq + 288b^2q^2 + 144bcq + 24bd + 9c^2\right) \\
& + 27c^2q^2 + 18cdq + d^2 \Bigg)
\end{aligned}
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\begin{aligned}
\text{Bias}(g_{\text{naive}})^2 = 4\frac{\Delta}{\epsilon}^4 \Bigg(& 100a^2q^6 + 120abq^5 + 60acq^4 + 20adq^3 + 36b^2q^4 + 36bcq^3 \\
& + 12bdq^2 + 144\frac{\Delta}{\epsilon}^4 \left(25a^2q^2 + 10abq + b^2\right) \\
& + 24\frac{\Delta}{\epsilon}^2 \left(50a^2q^4 + 40abq^3 + 15acq^2 + 5adq + 6b^2q^2 + 3bcq + bd\right) \\
& + 9c^2q^2 + 6cdq + d^2 \Bigg)
\end{aligned}
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